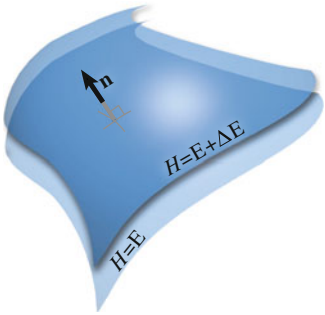


Fig. 5.1 A thickened shell between energy levels E and E + ΔE

Let us attempt to motivate the definition of the invariant measure. Consider a subset M of the level set Σ_{E}. We suppose $\|\nabla H\|$ to be bounded away from zero on Σ_{E}. At each point of M we let $\boldsymbol{n} = \nabla H/\|\nabla H\|$ be the unit normal to the energy surface and define a local coordinate ξ measuring distance along the direction of the unit normal, so that for a small offset $\Delta\mathrm{E}$ of the energy, $z + \xi\boldsymbol{n}$ is the point of intersection with the level set $\Sigma_{\mathrm{E}+\Delta\mathrm{E}}$. We thus obtain a thickened shell of points around the patch M with energy between E and E + ΔE (Fig. 5.1).

The volume of this thickened shell is clearly

$$\Delta\mathrm{V} = \int_{\mathcal{M}} \xi \mathrm{d}\sigma,$$

but since

$$\Delta\mathrm{E} = H(z + \boldsymbol{n}\xi) - H(z) = \nabla H \cdot \boldsymbol{n}\xi + \mathcal{O}(\Delta\mathrm{E}^2),$$

and $\boldsymbol{n} = \nabla H/\|\nabla H\|$, we have

$$\xi = \frac{\Delta\mathrm{E}}{\|\nabla H\|} + \mathcal{O}(\Delta\mathrm{E}^2),$$

and

$$\Delta\mathrm{V} = \int_{\mathcal{M}} \frac{\Delta\mathrm{E}}{\|\nabla H\|} \mathrm{d}\sigma + \mathcal{O}(\Delta\mathrm{E}^2).$$

Taking the ratio of ΔV to ΔE we find

$$\frac{\Delta\mathrm{V}}{\Delta\mathrm{E}} = \int_{\mathcal{M}} \frac{1}{\|\nabla H\|} \mathrm{d}\sigma + \mathcal{O}(\Delta\mathrm{E}).$$